



Cumulative benefits of nature: Childhood nature experiences and daily well-being in young adults

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ARTICLE INFO

Handling editor: L. McCunn

Keywords:

Childhood nature experiences
Well-being
Daily diary study
Nature connectedness

ABSTRACT

The positive impact of nature on well-being has been well-documented. Yet, the relationship between childhood nature experiences and well-being in young adults is not as well-understood. Utilizing a 14-day daily diary method, the present study was designed to examine the relationship between childhood nature experiences and well-being in adulthood, as well as the mediating role of nature connectedness. In Study 1, we found that childhood nature experiences positively predicted daily well-being in a sample of 240 young adults ($M_{\text{age}} = 19.48$ years, age range 16–27 years; 215 females), even after controlling for gender, age and socioeconomic status. In an independent sample ($N = 303$; $M_{\text{age}} = 19.45$ years, age range 17–29 years; 247 females), Study 2 replicated this result and extended the findings by controlling for other positive childhood experiences and identifying nature connectedness as a mediator of the association. Collectively, these results indicate that early experiences in nature bring about enduring benefits, where nature connectedness acts as a bridge to translate these benefits into greater well-being in adulthood. These findings advance understanding of the long-term psychological benefits of childhood nature experiences and further provide valuable insights for the intervention programs aimed at promoting well-being.

1. Introduction

Nature experiences are a ubiquitous element in human existence, constituting a fundamental component of individuals' lives (Goldy & Piff, 2020; Soga et al., 2023). Childhood nature experiences (CNEs) encompass both sensory and cognitive processing of the mere presence of natural elements as well as physical interaction with nature (Sugiyama et al., 2021). Alarming, children nowadays are suffering from a progressive decline in nature contact. This phenomenon, often referred to as “extinction of experience”, describes a progressive decline in nature contact compared to previous decades (Colléony et al., 2020; Gaston & Soga, 2020). Meanwhile, concerns have emerged about the adverse effects of a diminished exposure to nature, a concept known as “nature-deficit disorder” (Louv, 2005). This issue is particularly concerning for children, as they are navigating critical windows of mental development. Natural environments provide unique sensory qualities and afford a distinct range of leisure activities, including physical activity and social interaction, which together constitute an important

context for supporting children's development. Certainly, spending time in nature as a child has been linked to better mental health (Li et al., 2025; Maes et al., 2021) and a reduced risk of psychiatric illnesses (Engemann et al., 2019, 2020; Shanahan et al., 2016) in adulthood. Nonetheless, the role of CNEs in young adults' everyday well-being remains insufficiently understood. The primary aim of the current research was therefore to investigate how CNEs relate to well-being using a daily diary study.

1.1. CNEs and well-being

Well-being is an ancient topic extensively discussed in philosophy. In psychology, well-being refers to positive affect and ease in life (Diener, 1984), along with personal growth and self-actualization (Disabato et al., 2016; Ryan & Deci, 2001). Specifically, this definition of well-being comprehensively encompasses transient emotional experiences oriented toward the present and long-term pursuits that entail integrating the past, the present, as well as the near future (Baumeister

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et al., 2013). The biophilia hypothesis (Kellert & Wilson, 1995; Soga & Gaston, 2021) may shed light on the association between CNEs and well-being in young adults. The hypothesis posits that humans possess an innate affinity for nature, since the human mind developed in natural environments. Thus, natural surroundings can inherently have a profound effect on individuals' developmental process (Zare Sakhvidi et al., 2023). Consistent with this perspective, the short-term benefits of nature exposure for children's well-being have been well-documented. Prior research has revealed that children with increased time spent in natural settings tend to report greater well-being (Cui et al., 2025; Moula et al., 2023).

Beyond these short-term effects, the long-lasting benefits of CNEs to well-being may persist into individuals' later lives. First, individuals with frequent interaction with nature in childhood may be more likely to appreciate natural elements in their daily lives, thereby enabling them to embrace more positive feelings (Bratman et al., 2021). Second, nature may serve as a resource for children to support their optimal development. By providing a relaxing and comfortable setting, nature facilitates autonomous engagement, which promotes short-term emotional regulation and satisfies fundamental psychological needs (Van Heel et al., 2023). In the long term, such exposure supports cognitive development and neurodevelopment, thereby contributing to their well-being as they grow up. Third, increased exposure to nature may help children develop healthy lifestyles, such as forming pro-environmental habits (Jensen & Olsen, 2019; Rosa et al., 2018; Shekari et al., 2024; Stehl et al., 2024; Van Heezik et al., 2021), which can in turn allow them to derive benefits from these behaviors (Guo et al., 2026; Michael, 2024). In fact, research has shown that greater CNEs are associated with lower levels of neuroticism and higher levels of openness in adulthood (Snell et al., 2020), personality traits that are related to well-being. More direct evidence comes from both cross-sectional studies (Li et al., 2022; Pame la Pensini et al., 2016; Vitale et al., 2022; Windhorst & Williams, 2015) and longitudinal designs (Engemann et al., 2021), which have found a positive link between CNEs and well-being in adulthood. These findings situate contact with nature during childhood as a potentially important predictor of adult well-being in its own right.

Although research has deepened our comprehension of the relationship between CNEs and well-being, additional investigation is necessary to advance our knowledge in this area. Firstly, the aforementioned studies typically defined CNEs as passive exposure to nature, overlooking the active components of children's interaction with nature. Secondly, these studies relied on retrospective well-being assessments covering extended periods (e.g., one month). Considering that evaluations of well-being using retrospective reports over extended periods often depend on the most intense and recent memories, such retrospective data may be prone to memory biases (Fredrickson & Kahneman, 1993). Thirdly, while Li et al. (2022) explored the link between childhood exposure to green spaces and well-being in adulthood, our study instead targets young adults. As a group situated in the transition from adolescence to adulthood, young adults exhibit a set of core developmental features: the exploration of identity, self-centered orientations, instability in daily life, and the perception of multiple life possibilities (Schulenberg et al., 2004). Although well-being tends to improve over the course of young adulthood (Galambos et al., 2015), experiences in early life may have a meaningful impact on well-being during this developmental stage (Li et al., 2023). Therefore, investigating the association between CNEs and well-being in young adults is warranted, so that programs to enhance well-being among young adults with limited CNEs can be developed.

Consequently, the present study incorporated children's active engagement with nature into the framework. More importantly, we employed the daily diary approach to track participants' everyday experiences of well-being. This intensive longitudinal tracking method reduces recall bias by shortening the time interval between an experience and its reporting (Reis et al., 2014). It also achieves higher ecological validity by measuring daily experiences spontaneously and

continuously (Bolger et al., 2003). In summary, this study employed a daily diary method aimed at exploring the association between CNEs and daily well-being.

1.2. Mediation of nature connectedness

To better grasp how CNEs are related to well-being, we posited that nature connectedness may serve as a possible mediating mechanism. Nature connectedness refers to an individual's subjective feeling of closeness to the natural environment (Mayer & Frantz, 2004; Nisbet et al., 2009; Schultz, 2002; Tam, 2013). Based on the biophilia hypothesis, there exists an inextricable bond between nature and the human mind forged through evolutionary processes; this relationship may consequently be key to improving mental health. Aligning with this notion, recent research has highlighted that nature connectedness may serve as a fundamental pathway to sustain and enhance well-being (Arola et al., 2023; Martin et al., 2020; Mayer et al., 2009). Similarly, meta-analysis have robustly demonstrated that being connected to nature can foster well-being (Capaldi et al., 2014; Pritchard et al., 2020).

Given that during childhood, individuals acquire autonomous outdoor exploration abilities that simultaneously fulfill their curiosity and cultivate an interest in nature, researchers pointed out that it is during childhood that nature connectedness may be primarily shaped (Gong & Li, 2024). In line with this notion, evidence from prior studies found that CNEs served as a vital predictor of nature connectedness in adulthood (Barrable et al., 2024). In other words, individuals who frequently engaged with natural environments during childhood tend to develop stronger nature connectedness as they grow up. As such, CNEs may elevate nature connectedness and consequently facilitate well-being. Alternatively, the absence of CNEs may result in a lack of interest in nature. This detachment from nature may not only render individuals more vulnerable to negative emotions when spending time in natural environment (Soga et al., 2023), but also prevent them from achieving self-actualization and experiencing personal growth through a harmonious coexistence with nature (Pritchard et al., 2020).

Collectively, a close association exists between CNEs, well-being and nature connectedness. Moreover, CNEs seem to contribute to nature connectedness, which further leads to high levels of well-being. Thus, it is rational to infer that nature connectedness functions as a mediator linking CNEs to well-being.

1.3. The current study

Our study made fourfold extensions to prior research. Firstly, we adopted Sugiyama's et al. (2021) definition of CNEs, which includes children's active engagement with nature, rather than merely passive exposure to nature. Secondly, the underlying mechanisms that link CNEs to well-being have not yet been fully explored. The present study examined whether nature connectedness acted as a mediator to better shed light on this link. Thirdly, given that prior research has established the long-lasting influence of other positive childhood experiences beyond nature experiences on well-being in adulthood (Cunha et al., 2024), we incorporated them in our model as a control variable to elucidate the unique contribution of CNEs. Finally, our studies used the daily diary method to capture the everyday experience of participants' nature connectedness and well-being. The intensive longitudinal tracking method ensures more objective and reliable data collection, thereby reducing retrospective biases compared with traditional retrospective measurement methods.

Taken together, the present study sought to explore how CNEs relate to daily well-being in adulthood, with attention to the mediating function of nature connectedness. Two sequential daily diary studies were designed to address the overarching research question. In Study 1, we sought to firstly establish the core association by examining whether CNEs predicted daily well-being in 240 young adults, with adding gender, age and socioeconomic status as covariates. In Study 2, we

sought to replicate these findings and unpack their underlying mechanism by examining whether the relationship is mediated by current nature connectedness using another sample of 303 young adults. Moreover, we controlled for positive childhood experiences in Study 2 to reveal the independent association between CNEs and well-being. Briefly, across these two studies, we explored the pure and robust relationship between CNEs and well-being by excluding a series of confounding variables, and we also tested whether nature connectedness could mediate the relation.

2. Study 1

2.1. Methods

2.1.1. Participants

In the winter semester of 2022, a total of 264 students were recruited via targeted convenience sampling (e.g., campus posters) from a local college community in China for a 14-day consecutive daily diary study. Participants in the current study were undergraduate and graduate students, forming a cohort of young adults across diverse academic disciplines and with origins spanning all regions of China. Consequently, this sample constituted a culturally homogeneous group embedded within the mainstream Chinese cultural context. Among these participants, the analysis sample excluded 24 participants, primarily for the following reasons: (a) they failed to complete any diary entries ($n = 10$), or (b) they participated in the diary procedure for fewer than ten days ($n = 14$). The subsequent analysis was based on 240 participants (215 females; $M_{\text{age}} = 19.48$ years, $SD_{\text{age}} = 2.18$ years, range 16–27 years).

2.1.2. Procedure

The Institutional Review Board (IRB) of the affiliated university granted approval for the study protocol. All participants underwent the following study procedure. First, Participants furnished written informed consent prior to enrollment. Subsequently, participants were asked to complete the Childhood Nature Experiences Scale and necessary demographic variables via an online survey platform (www.wjx.cn).

During the subsequent 14 consecutive days of our daily surveys, we assessed participants' well-being for that day by sending them an online questionnaire link at 6 p.m. once a day. The link expired at midnight. Those unable to complete the questionnaire by the deadline received a notification at 9 a.m. the following day, and the timeframe of the questions was adjusted from "today" to "yesterday". Those who had not submitted the questionnaire by noon on the second day were marked as missing data for that day. Each participant received 30 RMB as compensation for their engagement in the experiment.

Notably, 12 of these 240 participants failed to complete the questionnaire for 1–2 days, resulting in a total of 14 missing diary entries. Given that each participant was required to complete 14 consecutive days of diaries, we obtained 3346 valid diary entries ($240 \times 14 - 14 = 3346$ entries) in total. Using the Monte Carlo simulation in Mplus, we calculated the least number of participants required to attain sufficient power (Muthén & Muthén, 2017). Results indicated that the design achieved 80.4% statistical power to detect a small-to-medium level-2 effect ($\beta = 0.20$) at an alpha level of 0.05, with 14 repeated observations per participant and a total sample of 200 individuals. Thus, our sample size proved sufficient to meet the target level of statistical power.

2.1.3. Measures

2.1.3.1. Within-person measures. Daily well-being was assessed using several well-being measures. Firstly, two items from the Satisfaction With Life Scale (SWLS) were picked based on factor loadings identified in prior research (Diener et al., 1985). The daily SWLS in Chinese version demonstrated adequate measurement properties (Li et al., 2023; Wang

et al., 2025). Participants rated each item on a 7-point Likert scale, ranging from 1 (*strongly disagree*) to 7 (*strongly agree*), higher total scores indicated a higher degree of life satisfaction. In our study, internal consistency was deemed satisfactory ($\omega = 0.844$ and 0.976 for within- and between-person level, respectively). Second, the Scale of Positive and Negative Experience (SPANE, Diener et al., 2010), measured on a 5-point Likert scale (1 = *never* to 5 = *always*), was employed to evaluate the affective facet of well-being. This scale consists of 12 emotional terms and its measurement properties have been well-established in the Chinese population (Guo et al., 2026; Zhang et al., 2023). The within-person omega reliability coefficients of these measures were 0.920 for positive affect and 0.855 for negative affect, and the between-person omega reliability coefficients were 0.992 for positive affect and 0.979 for negative affect. Finally, Keyes' (2009) 6-item psychological well-being subscale, part of the Mental Health Continuum-Short Form (MHC-SF), was utilized to measure the eudaimonic aspect of daily well-being. Studies have shown that the daily adaptation of the scale possesses adequate reliability and validity (Li et al., 2023; Yan et al., 2022). Participants rated each item on a 7-point Likert scale (1 = *strongly disagree* to 7 = *strongly agree*). The current study revealed adequate internal consistency for this scale ($\omega = 0.864$ and 0.977 for within- and between-person level, respectively).

Overall well-being was calculated through the following procedure. The SPANE was first converted to a 7-point scale. Item-averaged scores were then computed individually for life satisfaction, positive affect, negative affect and psychological well-being. Finally, we averaged the scores across these four subdimensions to derive the composite score. This score has been widely employed as an indicator of well-being in previous research (Joshi, 2011; Yang et al., 2025).

2.1.3.2. Between-person measures. Childhood nature experiences. To assess CNEs, participants were asked to report the frequency of their recreational activities in natural settings (i.e., parks, forests, farmland, and rivers/ocean) and nature-related engagements (including hiking, fishing, picking wildflowers and fruits) before age 12 (Sugiyama et al., 2021). A 5-point Likert scale was applied, with 1 representing "never" and 5 representing "very often". In the current study, the scale exhibited favorable reliability (Cronbach's $\alpha = 0.76$).

2.1.3.3. Control variables. The demographic variables including gender, age and socioeconomic status (SES) were included in our analysis as the control variables to rule out their influences, given established empirical evidence linking these variables to well-being (Charles & Piazza, 2009; Curhan et al., 2014; López-Madrugal et al., 2021; Präg et al., 2016; Yan et al., 2021). The MacArthur Scale of Subjective Social Status (Adler et al., 2000) was employed in our study to measure current SES.

2.1.4. Data analysis

Following Nezlek's (2012) methodological recommendations, we constructed a multilevel model incorporating both between-person and within-person level data. The analysis included a total of 3346 daily diary entries (Level 1) nested within 240 participants (Level 2). We conducted our data analysis in R 4.4.3 (R Core Team, 2025). Between-level descriptive statistics were computed using *bruceR* version 2024.6 (Bao, 2021) and the statistical models were estimated using Mplus 8.3 (Muthén & Muthén, 2017) through *MplusAutomation* 1.1.1 (Hallquist & Wiley, 2018).

The model construction proceeded in three stages. Firstly, a null model was established to assess the descriptive statistics and the intra-class correlation coefficients (ICCs). Next, we executed a multilevel regression model to test whether CNEs positively predicted well-being. In addition, the demographic variables were included as covariates. In this set of variables, daily well-being was classified as a within-person (level 1) variable, whereas CNEs and those covariates were categorized as between-person level (level 2) variables. Grand-mean centering

was applied to all between-person variables. Missing data were uniformly coded to 999.

2.2. Results

2.2.1. Preliminary analyses

Table 1 presents the descriptive statistics, ICCs and correlations at the between-person level for the focal variables. As illustrated, the ICC for daily well-being was 0.579, indicating that for this measure, 57.9% of the variance could be attributable to differences across individuals. Accordingly, the use of multilevel analyses is justified (Sonnetag et al., 2020).

2.2.2. The multilevel regression analysis

A multilevel regression model was constructed to examine whether CNEs predict well-being. As hypothesized, results indicated that CNEs positively predicted well-being ($\beta = 0.229$, $p = 0.001$, 95% CI = [0.095, 0.363]), after controlling for gender, age and SES, the result remained significant ($\beta = 0.198$, $p = 0.003$, 95% CI = [0.066, 0.329]). These findings indicated that CNEs exert a robust and meaningful predictive effect on well-being beyond the influence of SES, age, and gender.

3. Study 2

3.1. Methods

3.1.1. Participants and procedure

The study's protocol was approved by the Institutional Review Board (IRB) of the affiliated university. We recruited participants from a local college community via targeted convenience sampling during the winter of 2024 and obtained informed consent. The participants in the current study shared analogous characteristics with those of Study 1. They underwent a similar questionnaire collection process as in Study 1. A total of 318 young adults participated in the baseline measurement. Among them, 12 participants did not complete any diary entries and 3 participants finished the diaries less than ten days. Therefore, we removed 15 participants, ending up with a final number of 303 participants being included in the subsequent analysis (247 females, $M_{\text{age}} = 19.45$ years, $SD_{\text{age}} = 1.87$ years, range 17–29 years).

After the study was completed, the dataset lacked 25 days' entries due to insufficient participant responses. Consequently, we obtained 4217 daily diary entries in total, averaging 13.91 days per participant.

3.1.2. Measures

3.1.2.1. Between-person measures. Childhood nature experiences. The same measurement was used as in Study 1 to assess CNEs. The Cronbach's alpha was 0.82 for this scale.

3.1.2.2. Within-person measures. Nature connectedness. The Inclusion of Nature in Self (INS; Schultz, 2002) was employed to indicate participants' nature connectedness. This single-item measurement consists of a sequence of two circles signifying "self" and "nature" respectively, with depictions scaling from 1 (completely separate circles) to 7 (nearly total

overlap). Participants were required to pick one of the circles that most precisely reflected their relationship with nature. The degree of overlap between two circles represents participants' sense of nature connectedness. ICC(2) (Intraclass Correlation Coefficient Type 2) was used to examine the reliability of individuals' average nature connectedness across diary entries (Lüdtke & Trautwein, 2007), with the current sample showing a high reliability coefficient of 0.954.

Well-being. The measurement of well-being, including four subscales, was identical to that applied in Study 1. In Study 2, the omega reliability coefficients for these scales were highly acceptable. At the between-person level, the omega reliability ranged from 0.818 to 0.894. At the within-person level, the omega reliability ranged from 0.958 to 0.991. The well-being score was calculated using the same procedure as in Study 1.

3.1.2.3. Controlling variables. Consistent with Study 1, the control variables in this study included gender, age and SES. Besides, considering that more recent research has found other positive childhood experiences can exert an enduring influence on well-being (Wang et al., 2025), we used the Positive Childhood Experiences Scale to assess other positive childhood experiences (Bethell et al., 2019). Previous studies have verified the satisfactory psychometric properties of the scale in Chinese cultural contexts (Liu et al., 2025; Liu et al., 2025b). The scale exhibited satisfactory reliability in current study (Cronbach's $\alpha = 0.90$).

3.1.3. Data analysis

Following the methodological recommendations made by Nezlek (2012), we conducted multilevel analyses using data from daily entries (Level 1, $N_1 = 4217$ observations) embedded within individuals (Level 2, $N_2 = 303$ participants). First, a null model and a multilevel regression model were constructed. Then, a 2-1-1 mediation analysis was implemented, with nature connectedness as the mediator. Additionally, gender, age, SES and other positive childhood experiences were incorporated as the control variables. Other relevant settings were set to be consistent with those in Study 1 by default.

3.2. Results

3.2.1. Preliminary results

Table 2 presents an overview of means, standard deviations, ICCs, and correlations at the between-level among study variables. As predicted, well-being and nature connectedness were positively related to each other, and both were positively correlated with CNEs. Additionally, ICCs for nature connectedness (0.599) and well-being (0.560) fell within an appropriate range, thus warranting the use of multilevel analysis.

3.2.2. The multilevel regression analysis

Using multilevel regression modeling, we tested the relationship between CNEs and well-being. The analysis revealed that CNEs significantly predicted well-being ($\beta = 0.262$, $p < 0.001$, 95% CI = [0.157, 0.367]). Besides, after controlling for gender, age, SES and other positive childhood experiences, the positive relationship between CNEs and well-being remained significant ($\beta = 0.202$, $p = 0.002$, 95% CI = [0.076, 0.328]), suggesting the robustness of this link.

Table 1

Descriptive statistics and correlations among key variables.

Variables	<i>M</i>	<i>SD_b</i>	<i>SD_w</i>	ICC	1	2	3	4	5
1. Gender	-	-	-	-	1				
2. Age	19.475	2.177	-	-	-0.114	1			
3. SES	4.450	1.401	-	-	-0.027	-0.073	1		
4. CNEs	2.678	0.647	-	-	-0.057	-0.033	0.096	1	
5. WB	3.150	0.750	0.640	0.579	-0.059	-0.041	0.334***	0.229**	1

Note. *SD_w* and *SD_b* represent within- and between-subject level standard deviation; SES, socioeconomic status; CNEs, childhood nature experiences; WB, well-being.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table 2

Descriptive statistics and correlations among key variables.

Variables	<i>M</i>	<i>SD_b</i>	<i>SD_w</i>	ICC	1	2	3	4	5	6	7
1. Gender	-	-	-	-	1						
2. Age	19.449	1.868	-	-	-0.104	1					
3. SES	4.710	1.613	-	-	-0.112	0.240***	1				
4. PCEs	3.765	0.829	-	-	-0.025	0.028	0.252***	1			
5. CNEs	2.790	0.744	-	-	-0.061	0.063	0.196***	0.370***	1		
6. NC	4.272	1.226	1.003	0.599	0.096	0.057	0.121*	0.110	0.196***	1	
7. WB	3.282	0.728	0.646	0.560	0.041	0.074	0.107	0.236***	0.261***	0.462***	1

Note. *SD_w* and *SD_b* represent within- and between-subject level standard deviation; SES, socioeconomic status; PCEs, positive childhood experiences; CNEs, childhood nature experiences; NC, Nature Connectedness; WB, well-being.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

3.2.3. The multilevel mediation analysis

We constructed a multilevel mediation model to test the potential mediating effect of nature connectedness, with adding gender, age, SES and other positive childhood experiences as covariates. As anticipated, results showed a positive relationship between CNEs and nature connectedness ($\beta = 0.196$, $p < 0.001$, 95% CI = [0.088, 0.303]). Furthermore, nature connectedness demonstrated strong positive associations with well-being ($\beta = 0.409$, $p < 0.001$, 95% CI = [0.294, 0.524]).

As illustrated in Fig. 1, CNEs were significantly associated with well-being through nature connectedness (indirect effect = 0.077, $p = 0.003$, 95% CI = [0.027, 0.128]). Notably, with nature connectedness incorporated as mediator, the direct effect of CNEs on well-being remained significant ($\beta = 0.126$, $p = 0.030$, 95% CI [0.012, 0.241]).

4. Discussion

Using a daily diary design, we carried out two studies to test the association of CNEs with well-being in young adults and further explore the mediating role of nature connectedness. Our results revealed that CNEs significantly predicted well-being. This finding remained robust across two studies, even when gender, age, SES and other positive childhood experiences were controlled for. Furthermore, nature connectedness acted as a mediator in explaining the association between CNEs and well-being. Notably, the present work is the pioneer in employing the daily diary approach to explore the association as well as the mechanism between CNEs and well-being. This method can provide a more objective and reliable assessment of average nature connectedness and well-being levels over a period of time compared to traditional

questionnaire-based methods, thus making our findings more reliable. Taken together, these results offered preliminary evidence that CNEs are associated with daily well-being among young adults through nature connectedness.

4.1. The long-term predictive role of CNEs in well-being

In light of the various benefits of nature exposure in childhood (e.g., green space exposure and blue space exposure) on personal development, as noted earlier, the present study found that children with frequent experiences in nature reported significantly higher levels of daily well-being in their later lives. The biophilia hypothesis may account for this pattern of results (Kellert & Wilson, 1995). The hypothesis posited that humans have an inborn attachment to nature over the long history of evolution. As such, actualizing this innate affiliation, such as spending more time interacting with nature and being exposed to nature in childhood, can induce flourishing and be beneficial for child development. Fitting well with the hypothesis, prior research has verified the wide-ranging psychological benefits of nature contact. Beyond the restorative functions outlighted in classic theories, such as stress recovery and attention restoration (Kaplan, 1995; Ulrich et al., 1991), nature contact is also linked to healthier social bonds (Arbuthnott, 2023), more positive body image (Swami et al., 2020), increased authenticity (Yang et al., 2024), and a wide range of other positive outcomes (e.g., Bratman et al., 2021; Cheng et al., 2024; Weinstein et al., 2009; Yang et al., 2022). Moreover, research further suggested that exposure to nature in individuals' early life may cultivate sustained mental health benefits that persist into their later life (Li et al., 2025; Maes et al., 2021; Preuß et al., 2019). Importantly, our studies capture

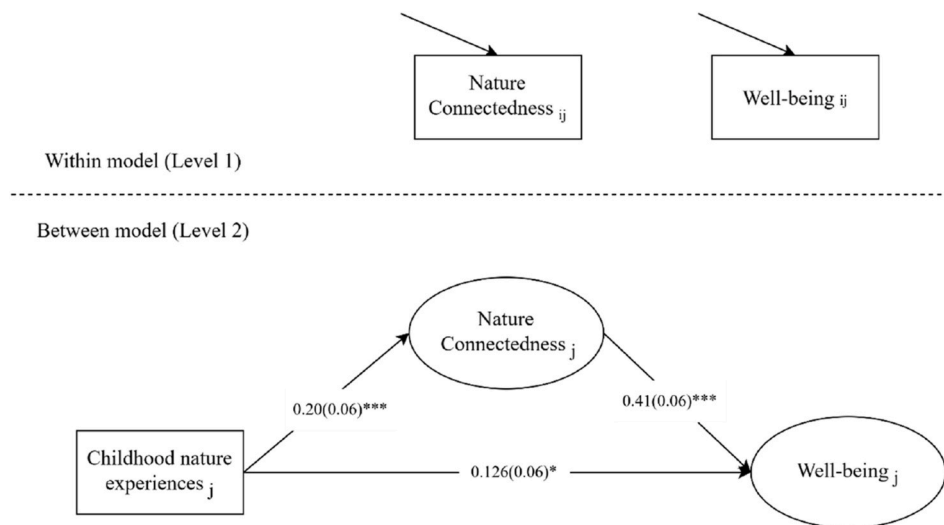


Fig. 1. The multilevel structural equation model involving CNEs, nature connectedness, and well-being in a 2-1-1 design ($N_1 = 4217$ observations; $N_2 = 303$ participants). *i* indexes individual cases; *j* indicates cluster; Standardized multilevel regression coefficients are presented: β (SE), * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

the enduring benefits of both children's passive exposure to natural settings and their active engagement in nature-related activities, rather than focusing solely on passive exposure to green spaces (Engemann et al., 2019; Li et al., 2022, 2025) or blue spaces (Engemann et al., 2021; White et al., 2020). Our findings further indicate that CNEs confer benefits for well-being, contributing to the growing literature on nature and positive psychological outcomes.

4.2. The mediated role of nature connectedness

Beyond establishing the link between CNEs and well-being, we further identified that nature connectedness serves as an underlying mechanism mediating this relationship, thus offering a deeper understanding of the ties between CNEs and well-being. These results align with prior research, which has convergently revealed that early-life communication with nature is conducive to higher nature connectedness in adulthood (Barrable et al., 2024; Gong & Li, 2024; Stehl et al., 2024). Concretely, CNEs can foster individuals' relationships with nature and help them integrate nature more deeply into their self-concept. Subsequently, the close connection to nature may in turn promote their levels of well-being. Self-Determination Theory provides a framework for understanding the effects of nature connectedness on well-being, proposing that human flourishing is predicated on the satisfaction of three basic psychological needs: autonomy, competence, and relatedness (Ryan & Deci, 2000; Vansteenkiste et al., 2020). Prior research has suggested that nature connectedness may represent a broad form of relatedness that supports well-being. Evidence further indicates that its association with well-being remains robust even after accounting for other social and cultural connections (Zelenski & Nisbet, 2014), highlighting the possibility that nature connectedness contributes uniquely to well-being beyond traditional forms of relatedness. Although some studies have suggested that nature connectedness may also give rise to environmental concern (Doherty & Clayton, 2011; Tam, 2013), which in turn may undermine individual's well-being. However, such concern is typically triggered by specific environmental events and may not generalize to individuals' actual daily experiences. By employing a daily diary design with repeated daily assessments, our study enhances ecological validity, reduces recall bias arising from recent episodes of memory (Fredrickson & Kahneman, 1993), and more accurately captures the relationship between nature connectedness and well-being as it unfolds in everyday life.

In addition, prior research has highlighted the enduring ramifications of positive childhood experiences on well-being in adulthood (Cunha et al., 2024; Wang et al., 2025), but these studies typically defined positive childhood experiences as nurturing relational and environmental factors including secure parent-child attachment, positive parenting practices, and supportive relationships with peers, school, and community. It is noticeable that CNEs can be considered a distinct type of positive childhood experience, different from the aforementioned other positive childhood experiences. Therefore, when examining the relationship between CNEs and well-being, it is crucial to account for the impact of other positive childhood experiences to isolate CNEs' unique contribution. In this study, we initially replicated the established correlation between other positive childhood experiences and well-being, and subsequently identified the unique link of CNEs with well-being after ruling out the plausible confounding effect of other positive childhood experiences. Additionally, the mediating role of nature connectedness also stayed significant after controlling for other positive childhood experiences. Taken together, these findings suggest the specific role of CNEs in human well-being.

4.3. Limitations

Several limitations warrant consideration, despite the significant findings of this study. First, the current study's sample was recruited from Chinese young adults attending a local public college. Future

research should examine the extent to which these findings can be generalized to other heterogeneous populations. Second, the present research primarily relied on self-report measures for data collection. Future research should include other assessment approaches in their study, such as interviews, to complement the self-report data. Thirdly, our discovery sample was characterized by a low proportion of male participants, which may affect the generalizability of the current findings. Subsequent studies are advised to address this imbalance by equalizing the gender distribution. Fourthly, we employed the single-item Inclusion of Nature in Self (INS) scale to assess daily nature connectedness. While this choice was grounded in pragmatic considerations to minimize participant response burden and ensure participant response quality within our diary study design, future research should develop and validate brief, multi-item scales specifically tailored to the assessment of daily nature connectedness in diary or experience sampling protocols, so as to further enhance measurement rigor.

4.4. Implications

The present findings offer meaningful theoretical implications. Specifically, we adopted a daily diary approach to establish the link between CNEs and subsequent well-being in adulthood, offering new empirical support for the biophilia hypothesis. Meanwhile, we identified the mediating role of nature connectedness, underscoring both the role of CNEs in fostering nature connectedness and the importance of nature connectedness for well-being. Moreover, by accounting for other positive childhood experiences, we demonstrated the unique association between CNEs and well-being. Collectively, this work offers novel insights into the enduring psychological benefits of CNEs.

As "nature-deficit disorder" intensifies amid the growing alienation of humans from the natural world, our empirical findings also offer practical insights across diverse fields. For urban planning, governments should prioritize green space development in urban design, particularly in residential areas and school districts. Within the context of the education sector, schools should integrate regular nature immersion into curricula, such as adding outdoor classroom programs and introducing dedicated modules focused on nature-related knowledge. Moreover, parenting guidelines should formalize nature exposure as essential to child development, encouraging structured green routines and unstructured nature play. Last but not least, in light of the mediating role of nature connectedness in linking CNEs to well-being, individuals affected by "nature-deficit disorder" can adopt actionable strategies to counter the ill effects of living deprived of nature during childhood. These strategies include increasing current nature contact (Martin et al., 2020) and writing gratitude letters to nature (Jacobs & McConnell, 2024), according to recent research.

5. Conclusion

To our knowledge, this is the first study investigating the psychological mechanism connecting CNEs with well-being in adulthood through a daily diary approach. Our findings show that CNEs are positively linked to daily well-being in adulthood, with nature connectedness functioning as a mediator. Moreover, by ruling out the effects of other positive childhood experiences, the findings reveal that CNEs represent a unique pathway to well-being, distinct from other positive childhood experiences. To sum up, this study offers new insights into the environmental psychology literature and highlights the importance of nature connectedness.

CRediT authorship contribution statement

Qian Liu: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization. **Shu Wang:** Writing – review & editing, Writing – original draft. **Xinchen Niu:** Writing – review & editing, Software. **Chengcheng Li:** Writing – review & editing,

Methodology. Tiantian Guo: Writing – review & editing, Software. Xuanqi Chen: Resources, Data curation. Feng Kong: Writing – review & editing, Validation, Project administration, Investigation, Funding acquisition.

Ethics statement

This study received approval from the Ethics Committee of the School of Psychology at Shaanxi Normal University. Informed consent was obtained from all participants involved in the study.

Funding

This work was supported by the National Natural Science Foundation of China [32371121], Fundamental Research Funds for the Central Universities (GK202501006) and ‘111’ center (B25068).

Declaration of conflicting interest

The authors have no relevant financial or non-financial interests to disclose.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jenvp.2026.103076>.

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